

ASSIGNMENT-1

MODULE-1

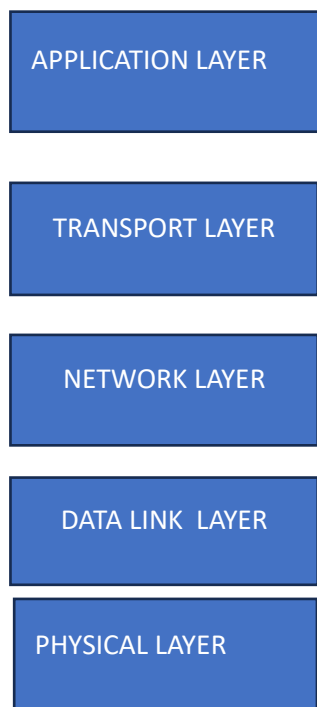
COMPUTER NETWORKS

(PCCST501)

Assignment Title: "How the Internet Delivers a Web Page"

Website choosen: <https://www.amazon.in>

1. Draw the TCP/IP protocol stack and explain the role of each layer.



>1. Application Layer

The **Application Layer** provides network services directly to user applications and acts as the interface between the user and the network. It enables services such as web browsing, email, file transfer, and remote login. Common protocols include **HTTP, FTP, SMTP, and DNS**.

2. Transport Layer

The **Transport Layer** ensures reliable end-to-end communication between devices. It divides data into segments, performs error checking, controls data flow, and reassembles data at the destination. The main protocols used are **TCP** and **UDP**.

3. Network Layer

The **Network Layer** is responsible for routing data packets from the source to the destination across different networks. It provides logical addressing using **IP addresses** and selects the best path for data transmission. Common protocols include **IP** and **ICMP**.

4. Data Link Layer

The **Data Link Layer** provides reliable communication between two directly connected devices on the same network. It converts packets into frames, uses **MAC addresses** for device identification, and performs error detection and flow control. Technologies such as **Ethernet** and **Wi-Fi** operate at this layer.

5. Physical Layer

The **Physical Layer** is responsible for transmitting raw bits over the physical communication medium. It defines the electrical, mechanical, and signaling characteristics of the network, including cables, connectors, and wireless signals. Examples include **Ethernet cables, fiber optics, and radio waves**.

2. Explain the role of DNS in converting the domain name into an IP address.

>DNS (Domain Name System) is a service that converts a human-readable domain name into an IP address. Since computers communicate using IP addresses, DNS helps users access websites without memorizing numerical addresses.

Example: Accessing Amazon

Suppose you want to visit www.amazon.com

1. User enters the domain name
 - You type www.amazon.com into your web browser.
2. Browser checks its cache
 - The browser first checks whether it already knows the IP address of www.amazon.com.
 - If it is not found, it sends a request to a DNS resolver.
3. DNS lookup
 - The DNS resolver queries:
 - Root DNS Server → directs the request to the .com TLD server.
 - .com TLD Server → points to Amazon's authoritative DNS server.

- Amazon's Authoritative DNS Server → returns the IP address of www.amazon.com.

4. IP address is returned

- DNS returns an IP address (for example, 54.239.28.85).
Note: Amazon uses many servers, so the IP address may vary depending on your location and load balancing.

5. Website loads

- The browser uses the IP address to connect to the Amazon server, and the Amazon homepage is displayed.

3. Describe how HTTP is used to request and receive the webpage.

>HTTP (Hypertext Transfer Protocol) is the communication protocol used by web browsers and web servers to transfer web pages over the Internet. When a user enters a website address, the browser sends an HTTP request to the web server. The server processes the request and returns an HTTP response containing the requested webpage (HTML, images, CSS, etc.). The browser then displays the webpage to the user.

Steps:

1. The user enters a URL in the web browser.
2. The browser sends an HTTP request to the web server.
3. The web server processes the request and sends an HTTP response with the requested webpage.
4. The browser receives the response and displays the webpage.

User → Browser → HTTP Request → Web Server

User ← Browser ← HTTP Response ← Web Server

4. Explain how TCP provides reliable communication for HTTP.

TCP (Transmission Control Protocol) provides reliable communication for HTTP by ensuring that data is delivered accurately and in the correct order. Before transmitting data, TCP establishes a connection between the client

and the server using a **three-way handshake**. It divides data into segments, assigns sequence numbers, detects errors, retransmits lost packets, and acknowledges received data. These features ensure that web pages are delivered completely and without errors.

5. Identify whether the application follows a Client–Server or Peer-to-Peer architecture and justify your answer.

A web application follows the **Client–Server architecture**. In this model, the **client** (web browser) sends requests for web pages, and the **server** processes those requests and sends the requested data back to the client. The communication is centralized because all services and resources are managed by the server, making it different from a Peer-to-Peer (P2P) network where devices communicate directly with each other.

6. Compare HTTP and FTP with at least four differences.

HTTP	FTP
HTTP stands for Hypertext Transfer Protocol .	FTP stands for File Transfer Protocol .
Used to access and transfer web pages.	Used to upload and download files between computers.
Uses port 80 (HTTP) and 443(HTTPS)	Uses port 21 for control and port 20 for data transfer
Transfers HTML pages, images, CSS, and web content.	Transfers files such as documents, images, videos, and software.
Does not require authentication for most public websites.	Usually requires a username and password for access.

7. Explain how cookies improve user experience with one real-world example.

>How Cookies Improve User Experience

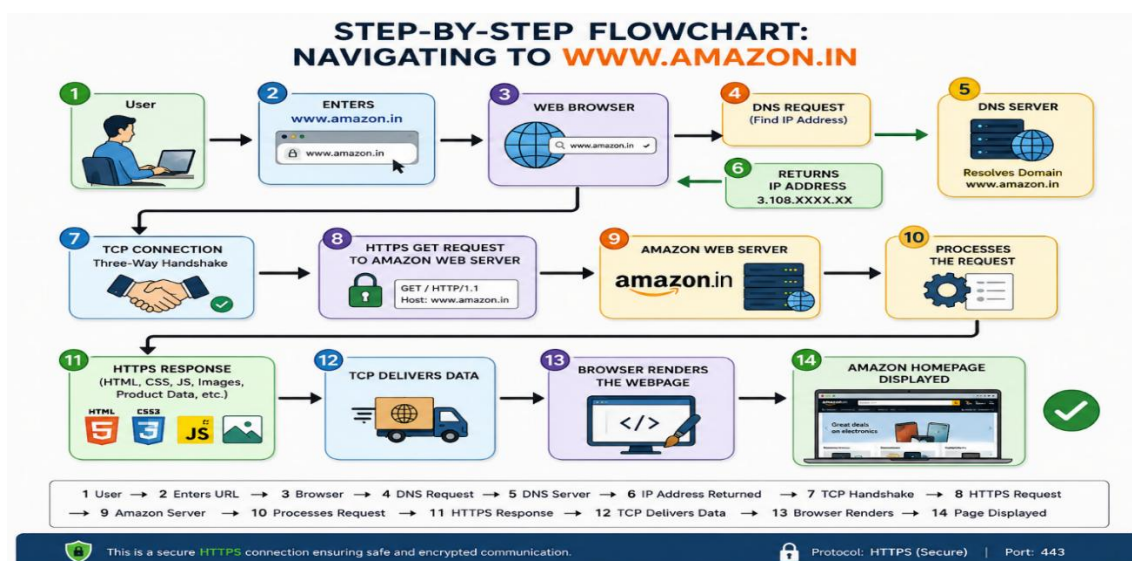
- **Remember Login Information:** Cookies keep users logged in, so they do not have to enter their username and password every time they visit the website.
- **Store User Preferences:** They remember settings such as language, theme (light/dark mode), and location preferences.
- **Maintain Shopping Cart:** Cookies save the items added to an online shopping cart, even if the user closes the browser and returns later.

Personalize Content: Websites use cookies to display recommendations, advertisements, and content based on the user's interests and previous browsing history.

- **Real-World Example: Amazon**

- When a user visits **Amazon** and logs into their account, the website stores a cookie in the browser. This cookie remembers the user's login session, language preference, and the items added to the shopping cart. If the user closes the browser and visits Amazon again later, they usually remain logged in, and the shopping cart still contains the previously selected items. Amazon also uses cookies to recommend products based on the user's browsing and purchase history, making shopping faster and more personalized.

8. Draw a flow diagram showing the complete communication from the browser to the web server and back.



9. Write a brief conclusion (100–150 words) describing the importance of application-layer protocols in everyday Internet usage.

Application-layer protocols are essential for enabling communication between users and Internet services. They define the rules that allow applications such as web browsers, email clients, file transfer programs, and messaging apps to exchange data efficiently. Protocols like **HTTP/HTTPS** support web browsing, **SMTP** and **IMAP/POP3** handle email communication, **FTP** enables file transfers, and **DNS** translates domain names into IP addresses, making websites easy to access. These protocols work together to ensure secure, reliable, and efficient communication across the Internet. Without application-layer protocols, everyday online activities such as browsing websites, sending emails, streaming videos, online shopping, and using cloud services would not be possible. Therefore, application-layer protocols form the foundation of modern Internet communication, providing users with a seamless and user-friendly online experience.